Automated Ambient Light Control and Actuation System for Smart Environments

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Abstract:

In both personal homes and large-scale corporate or educational facilities, varying ambient light levels and the mismanagement of the environmental controls results in compromised occupant comfort and significant energy waste. This paper proposes a localized, edge-computing Internet of Things (IoT) system designed to autonomously regulate physical window shutters, artificial lighting, and heating and AC systems. Utilizing a sensor fusion approach and Bluetooth Low Energy (BLE) for secure local configuration, the system prioritizes natural light utilization and actively terminates power to lighting and climate controls in unoccupied spaces, eliminating the long term passive power draw and higher costs associated with cloud-dependent architectures.

I-Introduction:

In residential homes or large corporate or educational buildings, changing ambient light and poor environmental control lead to both occupant discomfort and substantial energy waste.

The proposed architecture addresses the first challenge that is the varying light levels by autonomously adjusting heavy exterior window shutters and indoor lighting based on real-time ambient light readings. As external light intensity rises, the shutters are lowered to maintain a target light level and reduce solar heat gain. With the same logic, as external light decreases, the shutters are elevated. When ambient light drops below a critical limit where natural light is no longer sufficient even with fully open shutters, the system transitions to a "night state," deploying the shutters completely for thermal insulation and privacy, while engaging the indoor lighting. By relying strictly on closed-loop sensor feedback rather than an internal calendar, the system inherently adapts to shifting daylight hours throughout the year.

In addition to this, the integration of advanced motion sensing addresses the issue of energy waste in unoccupied spaces. The system actively monitors the room and will systematically trigger a shutdown sequence after a specified duration of zero-motion. This sequence cuts power to the artificial lighting and utilizes an infrared transmitter to power down existing localized heaters and AC units, all without interfering with manual user overrides. By isolating the system from internet access and relying entirely on localized edge processing and BLE for configuration, the architecture ensures higher reliability during network outages while remaining easily adjustable via a mobile application.

II-Proposed System Layout

A. Processing and Power Distribution:

The central processing unit is an ESP32[1] microcontroller, selected for its dual-core architecture. To eliminate control latency, Core 0 is dedicated to handling BLE communications for system configuration, while Core 1 executes the high-frequency control loops for sensors and motor interrupts. The hardware architecture utilizes a tiered power distribution system to safely isolate high-current electromechanical components from sensitive logic circuits. A dedicated step-down converter provides a stable 3.3V and 5V logic supply to the ESP32 and sensor modules, while a separate high-current power tier is provisioned strictly for the actuator loads.

B. Sensor Management:

To mitigate false positives in occupancy detection, a sensor fusion approach is used. An HC-SR501[2] Passive Infrared (PIR) sensor acts as an instant hardware trigger for macro-movements (e.g., a person entering the room). Simultaneously, an HLK-LD2410[3] 24GHz millimeter-wave (mmWave) radar monitors sustained micro-movements to maintain the active occupied state. Ambient light is quantified using a BH1750[4] digital I2C sensor, bypassing the non-linearities and analog-to-digital conversion noise associated with traditional photoresistors.

C. Actuation:

Physical actuation is separated into three outputs. Artificial lighting is controlled by a standard 5V relay module. The window shading mechanism, consisting of heavy exterior roller shutters, requires high torque and is driven by a heavy-duty electromechanical actuator. This motor is controlled by a dual-relay configuration to dictate absolute forward (roll up) and reverse (roll down) states, ensuring the microcontroller's logic levels remain safely isolated from the high power draw. Finally, localized climate control (HVAC) is managed via a 38kHz Infrared (IR) Transmitter module driven by the ESP32, allowing the system to interface safely with existing split-unit ACs and heaters without intercepting high-current mains power.

To preserve user autonomy, a traditional wall switch is integrated without compromising the system's power delivery. The switch does not directly break the mains circuit; instead, it is wired to trigger a GPIO interrupt on the ESP32. The microcontroller's firmware evaluates the change in state to toggle the light relay. This state-change logic allows for seamless transitions between automated occupancy routines and manual user overrides.

D. Simulation and Virtual Validation:

Due to physical deployment constraints, initial system validation and control logic verification will be conducted within a virtual simulation environment. This deployment allows for the rigorous testing of the ESP32 firmware, I2C sensor polling, and state-change interrupt logic prior to physical hardware assembly.

III. SYSTEM CONFIGURATION AND INTERFACE

To maintain edge-computing security and eliminate the need for an active internet connection, system parameters are managed locally via a BLE interface hosted on the ESP32 Core 0. The microcontroller exposes a secure GATT (Generic Attribute Profile) server that a companion mobile application can connect to for configuration.

Through this interface, users can adjust critical system variables stored in the microcontroller's non-volatile memory. These parameters include:

Target Light Level: The desired ambient light level the system actively attempts to maintain.

Predetermined Limit Steps: A defined tolerance range around the intended light level to prevent the shutter motors from continuously oscillating due to minor light fluctuations.

Override Duration: A programmable timer that dictates how long the system remains in manual mode after the physical wall switch is toggled.

Occupancy Timeout: The duration of zero-motion required from both the PIR and mmWave radar sensors before transitioning to an unoccupied energy-saving state.

Heater & AC Automation Toggle: A configurable boolean parameter (Enable/Disable) that dictates whether the system should transmit the IR shutdown signal to the AC and heating units upon reaching the occupancy timeout.

REFERENCES:

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